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IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

SPEECH PROCESSING

Friday, 16 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : W. Dai

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Speech Processing

1. Consider a lossless tube model of the vocal tract using two sections of tube. Figure 1.1 shows signal flow graphs of alternative representations of this lossless tube model labelled Structure A and Structure B.

   a) Derive an expression for $\begin{bmatrix} V_1^+ \\ V_1^- \end{bmatrix}$ in terms of $\begin{bmatrix} V_2^+ \\ V_2^- \end{bmatrix}$.

   Then derive a further expression for $\begin{bmatrix} W_1^+ \\ W_1^- \end{bmatrix}$ in terms of $\begin{bmatrix} W_2^+ \\ W_2^- \end{bmatrix}$

   [6]

   b) Derive the transfer functions

   $$\frac{Y(z)}{X(z)} \text{ and } \frac{Q(z)}{P(z)}$$

   and hence give the relationship between these two transfer functions.

   [6]

   c) Consider the computations involved in computing one output sample in these two structures. State the number of addition operations needed and the number of multiply operations needed to compute one output sample using Structure A. Repeat this question for Structure B. [4]

   d) Now consider the two-tube representation given in Structure B. Also consider a sequence of input samples $\{p(n)\}$ and a corresponding sequence of output samples $\{q(n)\}$.

   Write out the software programming code that computes the output sequence $\{q(n)\}$ given an input sequence $\{p(n)\}$. You may use the syntax of MATLAB or any other language or pseudo-code providing it is clearly set out.

   Hint: Use one FOR loop in your code to count through each sample.

   Note that marks will not be deducted for syntax errors in the code providing that the operations are described unambiguously. It is not necessary to show any variable declarations or intialization in the code providing they are clear from the context. [4]

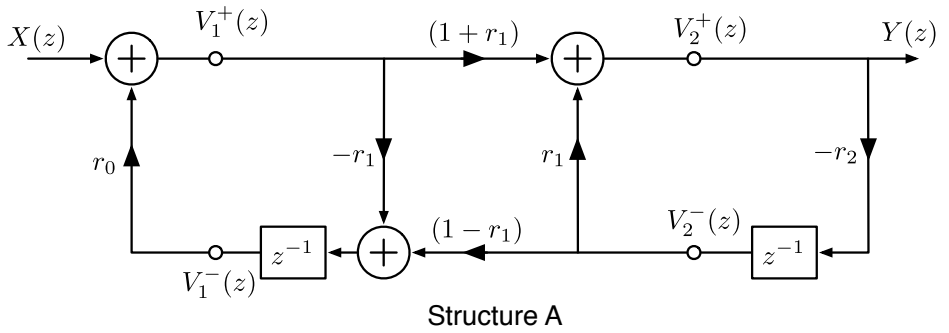


Figure 1.1: Structures representing a lossless tube model

2. Linear prediction is applied to a speech signal $s(n)$. The order of the LPC is chosen to be p. The resulting prediction coefficients are denoted a_k for $k = 1, \ldots, p$ and the LPC system function is given by

$$H(z) = \frac{1}{1 - \sum_{k=1}^{p} a_k z^{-k}}.$$

a) Why is LPC generally considered to be suitable for encoding speech signals? Include in your answer any relevant properties of speech signals. [4]

b) Consider (i) voiced and (ii) unvoiced speech. Briefly describe the characteristics of the prediction residual in the case when the prediction is exact. [2]

c) Summarise the method of prediction of pitch periodicity in LPC-based speech coding methods. [2]

d) Let i, k and m be indices. Consider a particular analysis frame containing a segment of the speech signal $s(n)$, and denote the corresponding prediction error $e(n)$. Now define

$$\varphi(i,k) = \sum_{m} s(m-i)s(m-k).$$

i) For this analysis frame, give the expression for the squared prediction error ε in terms of a_k and s. [2]

ii) Show how the prediction coefficients a_k can be chosen to minimize ε and write an expression for ε in terms of φ and a_k. [4]

e) Now consider a particular frame of speech for which the first 3 values of the autocorrelation function are 51, 45, 39. Determine the optimal linear prediction coefficients in $H(z)$ for 2nd order autocorrelation LPC and find the corresponding minimum squared error. [6]

3. a) A speech processing device for use in aircraft is being designed to include a noise reduction algorithm. The clean speech is denoted $s(n)$, the noisy speech is $y(n)$, and the noise is $\upsilon(n)$.

i) Give a brief summary of the operating principles of the spectral subtraction approach for noise reduction. Include a labelled block diagram of the spectral subtraction approach and a few sentences of explanation. *(Mathematical formulae or other analytical details are not required).* [3]

ii) Describe and explain the minimum statistics approach for noise estimation. Include relevant mathematical formulae in your description. Begin by defining the input and output signals of the minimum statistics approach. [5]

iii) Obtain an expression for the filter $H(l,k)$, for time frame l and frequency index k, that implements amplitude spectral subtraction exploiting a noise estimate. [4]

b) It is desired to perform a time-frequency analysis of a speech signal $s(n)$, with time index n, to give a spectrogram S.

i) Give the formula for the spectrogram S and define all variables and indices.

ii) Write 2 or 3 sentences to explain the processing employed in spectrographic analysis.

iii) Explain how the frequency resolution of spectrographic analysis is determined.

iv) Choosing a particular single frequency, explain the effect of any windowing applied. Support your explanation with any appropriate formulae and/or diagrams. [8]

4. a) When considering statistical methods of automatic speech recognition, explain why it is usually preferred to employ log probability densities instead of probability densities. [3]

b) A speech utterance is processed by a statistical speech recognition system using hidden Markov models. The speech signal is divided into $T = 5$ frames. The duration of each frame t is 20 ms. The set of frames is therefore written $\mathbf{x}_t$ for $t = 1, 2, \ldots, T$.

The hidden Markov model has $S = 4$ states, numbered 1, 2, 3 and 4. The output log probability densities for frame t in state i are $\log(d_i(\mathbf{x}_t))$ and are given in Table 1. The transition probability from state i to state j of the model is a_{ij} as shown in Fig. 4.1.

	Input Frame				
State	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$
1	-4	-8	-5	-5	-7
2	-2	-5	-7	-7	-6
3	-5	-3	-3	-8	-6
4	-6	-7	-8	-4	-8

Table 1: Output log probability densities

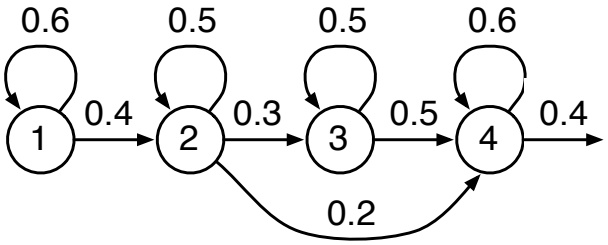


Figure 4.1: Hidden Markov model

i) For the case where

$$d_i(\mathbf{x}) = (2\pi)^{-1/2P}|\mathbf{C}_i|^{-1/2}\exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m}_i)^T\mathbf{C}_i^{-1}(\mathbf{x}-\mathbf{m}_i)\right)$$

write a simplified expression for $\log(d_i(\mathbf{x}))$. Explain any advantageous features in the computation involved in calculating $\log(d_i(\mathbf{x}))$. [4]

ii) The maximum probability density that the model generates frames $\mathbf{x}_1, \ldots, \mathbf{x}_t$ from any sequence of states for which frame 1 is in state 1 and frame t is in state s is defined as $B(t, s)$.

For the speech recognition task described above,

- write out a table showing the transition log probabilities from every state to every other state;
- draw the alignment lattice beginning with frame 1 in state 1, and ending with frame 5 in state 4;

- hence determine $\log(B(5,4))$ and the most likely state sequence.

[9]

c) Consider an alternative speech recognition system based on Deep Neural Networks (DNNs). Explain a common training strategy to improve the robustness of trained DNN models for speech recognition in real-world environments. [4]